Positivity and recurrence formulas for oscillatory Gegenbauer Hankel determinants

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Abstract

For $\alpha > -1$, consider the Hankel determinants of the complex weight $(1-t^2)^\alpha e^{ixt}$ on $[-1,1]$. We prove that every even-size determinant is strictly positive for every real frequency x. The argument extends the published exponent-zero theorem by continuation in α. Its two ingredients are a Pearson-identity proof that adjacent determinants never vanish together, and a large-frequency expansion uniform on compact exponent intervals. The first ingredient also implies that every complex zero of every determinant is simple. Odd-size determinants have infinitely many real zeros, and their first endpoint correction is computed explicitly. The same expansion proves two formal polynomial-degree conjectures stated in 2008 and gives complete proofs of two recurrence-coefficient conjectures stated in 2026. An earlier 2026 preprint claimed to derive those recurrence formulas; we show that its key determinant formula fails as printed, and our proofs do not use it. An exact Bessel–Jacobi determinant identity transfers the even-size result to spherical-Bessel derivative determinants. Combined with the published rank-three result (an unrefereed preprint), it proves the critical-length conjecture for the eight-dimensional space spanned by $x^j\cos x, x^j\sin x$, $0 \le j \le 3$. The general critical-length conjecture remains unresolved here.

Keywords. Hankel determinant; oscillatory orthogonality; Gegenbauer weight; Bessel function; critical length; endpoint asymptotics.

1 Introduction and main result

For $\alpha > -1$ and $z \in \mathbb{C}$, define

$$\mu_j^{(\alpha)}(z) = \int_{-1}^{1} t^j (1-t^2)^\alpha e^{izt}\,\mathrm{d}t, \qquad D_N^{(\alpha)}(z) = \det[\mu_{i+j}^{(\alpha)}(z)]_{i,j=0}^{N-1}, \qquad D_0^{(\alpha)} = 1. \tag{1.1}$$

For fixed real $\alpha > -1$, these are entire functions of the frequency. The determinants are real and even on the real axis. At frequency zero the measure is positive, so every determinant is positive. At nonzero frequency the measure is complex and ordinary positive-measure orthogonality no longer ensures that the determinants are nonzero.

Our principal statement is the following.

Theorem 1.1. *For every integer $r \ge 1$, every real $\alpha > -1$, and every $x \in \mathbb{R}$,*

$$D_{2r}^{(\alpha)}(x) > 0. \tag{1.2}$$

Consequently, the monic orthogonal polynomial of each even degree exists uniquely for $(1-t^2)^\alpha e^{ixt}$ on $[-1,1]$, at every real frequency.

Why this matters, in plain terms. Integrals of rapidly oscillating functions, such as $\int_{-1}^{1} f(t)(1-t^2)^{\alpha} e^{\mathrm{i}xt}\,\mathrm{d}t$, are expensive to compute by ordinary rules once the frequency x is large. One family of efficient methods builds Gaussian-type rules from polynomials that are "orthogonal" with respect to the complex weight itself. Because that weight is complex, such polynomials need not exist: the polynomial of degree N exists exactly when the Hankel determinant D_N is nonzero. Theorem 1.1 says that, for every Gegenbauer exponent and every real frequency, this never fails in even degree. It is an existence statement only; it does not by itself show that the resulting quadrature nodes are distinct or well conditioned. For this numerical context see Deaño and Huybrechs [6], Asheim, Deaño, Huybrechs and Wang [5], and the monograph [7].

How the paper is organised. Section 2 proves that two consecutive determinants never vanish together, so all zeros are simple. Section 3 derives large-frequency expansions with constants uniform in the exponent, reducing everything to one-variable endpoint integrals. Section 4 combines these to prove Theorem 1.1. Sections 5–7 use the same expansions for the 2008 degree conjectures and the 2026 recurrence conjectures. Section 8 records consequences and limits, Section 9 the checks and the priority record, and the appendix the Bessel–Jacobi identity. Readers interested only in the main theorem need Sections 2–4.

This does not assert existence of the entire orthogonal-polynomial sequence at every frequency. Each odd-size determinant has infinitely many positive real zeros; see Corollary 4.1. The distinction between even and odd degree is therefore necessary.

The case $\alpha = 0$ is the even-degree existence theorem of Celsus, Deaño, Huybrechs and Iserles [1, Theorem 4.3]. We use that published result as a base case. The extension to all $\alpha > -1$ follows from a continuation argument. First, a Pearson identity rules out simultaneous vanishing of adjacent determinants. The Toda identity then rules out multiple zeros. Second, a uniform endpoint expansion shows that an even-size determinant is positive at every sufficiently large real frequency, with a common threshold on each compact exponent interval. Real zeros consequently cannot be created during continuation, enter through frequency zero, or arrive from infinity.

The oscillatory Gegenbauer family was studied by Milovanović, Cvetković and Marjanović [2]. Their parameter is $\lambda = \alpha + \frac{1}{2}$. They prove existence of the entire sequence when $\lambda > -\frac{1}{2}$ is rational and the frequency is a nonzero zero of $J_{\lambda-1}$. They also conjecture degrees of two families of polynomial Hankel expressions. Lyu and Zhou [4] revisit these expressions, propose leading-coefficient products, and retain special parameter and frequency restrictions in their existence assumptions. Sections 5 and 6 prove both formal degree statements and give their leading coefficients. In the shifted case, the generic degree has two exceptional specialisations in the admissible real parameter interval.

Section 7 then proves all the formulas in [4, Conjectures 2.5 and 2.7], retaining their precise index ranges. Those conjectures assert that the recurrence coefficients are given by explicit rational expressions in certain monic polynomials X_j, Y_j of degree j, which Lyu and Zhou list only for low indices. What is proved here is therefore: an explicit construction of such polynomials for every index, the degree and leading-coefficient facts that make them monic of the stated degree, and the stated identities. Our polynomials coincide with those listed by Lyu and Zhou at the indices they give (Section 7).

Prior claim. A preprint by Ashfaque [16], posted on ResearchGate on 24 July 2026, claims that the Lyu–Zhou recurrence formulas follow from a discrete Painlevé IV and tau-function identification. It therefore precedes this paper as a public claim on Conjectures 2.5 and 2.7. As we explain in

Section 9, the determinant formula on which that derivation rests (its Proposition 3.1) fails as printed, so we regard the proofs here as the first complete ones known to us for those conjectures, not as the first claim. As far as our reading of that preprint goes, it does not state Theorem 1.1 or the 2008 degree results. These algebraic results use the endpoint asymptotics and determinant identities; they do not require Theorem 1.1. The positivity theorem additionally uses the exponent-zero base case and continuation.

The exact Fourier–Gegenbauer Wronskian object also occurs in Dimitrov and Xu [3]. Their positive-definiteness results for Fourier Wronskians are not pointwise positivity statements. Theorem 1.1 concerns pointwise positivity at every even size. An exact atomic example in Section 8 illustrates why positive definiteness, or symmetry of a positive underlying measure, does not itself prove this conclusion.

Jacobi-type kissing polynomials are also studied by Barhoumi [8] in a regime where degree and frequency grow together. Those large-degree asymptotic existence results have a different scope from the all-finite-even-degree, all-real-frequency assertion above.

An appendix supplies an exact identity with signed derivative Hankel determinants of spherical-Bessel functions. This identity is derived from the standard unitary orbital integral and the Jacobi law for random projection compressions; no separate novelty claim is made for that classical representation. It gives all even-size nonvanishing conditions in the critical-length problem of Carnicer, Mainar and Peña [10]. Combined with Kounchev and Render [11, Corollary 21], it settles the case $n = 3$ of that problem; this corollary is conditional on their result, which is an unrefereed preprint. Higher odd-size first-zero comparisons are not proved here.

The priority discussion in Section 9 records the bounded literature comparison, including an earlier competing claim. The mathematical statements below do not depend on treating a search result as proof of historical priority.

2 A Pearson identity and simple zeros

Fix a real $\alpha > -1$. For a polynomial p, write

$$L_z(p) = \int_{-1}^{1} p(t)(1-t^2)^{\alpha} e^{\mathrm{i}zt}\,\mathrm{d}t.$$

Integrating the derivative of $R(t)(1-t^2)^{\alpha+1}e^{\mathrm{i}zt}$ gives

$$L_z\Big((1-t^2)R' + [\mathrm{i}z(1-t^2) - 2(\alpha+1)t]R\Big) = 0. \tag{2.1}$$

The endpoint values vanish because $\alpha + 1 > 0$, and all terms are integrable. This identity holds for every polynomial R and every complex frequency.

Lemma 2.1. *There is no $z \in \mathbb{C} \setminus \{0\}$ and no integer $n \geq 1$ for which*

$$D_n^{(\alpha)}(z) = D_{n+1}^{(\alpha)}(z) = 0.$$

Proof. Fix a frequency and suppress α, z from the notation. If an adjacent pair vanishes, choose the least such n. Then $D_{n-1} \neq 0$; put $d = n - 1$. Nonsingularity of the size-d moment matrix provides a unique monic polynomial p of degree d satisfying

$$L(pq) = 0 \qquad (\deg q \leq d-1). \tag{2.2}$$

For $d = 0$, this means $p = 1$ and the condition is empty. The monic triangular basis change to $1, t, \dots, t^{d-1}, p$ gives $D_{d+1} = D_d L(p^2)$. Hence $L(p^2) = 0$, and p is orthogonal to every polynomial of degree at most d.

In the monic basis $1, t, \dots, t^{d-1}, p, tp$, the size-$(d+2)$ Gram matrix has the form

$$\begin{pmatrix} H_d & 0 & v \\ 0 & 0 & c \\ v^T & c & e \end{pmatrix}, \qquad c = L(tp^2).$$

The Schur complement of H_d therefore gives $D_{d+2} = -D_d c^2$. Since $D_{d+2} = 0$ and $D_d \neq 0$, we have $c = 0$. The polynomial tp is monic of degree $d+1$, so p is orthogonal to all polynomials of degree at most $d+1$.

Suppose inductively that this orthogonality holds through degree $M \geq d+1$. Set $k = M-d-1 \geq 0$ and substitute $R = t^k p^2$ in (2.1). Every term except $-\mathrm{i}zL(t^{k+2}p^2)$ has the form $L(pq)$ with $\deg q \leq M$. For example, the largest degree in the term involving p' is $\deg(t^{k+2}p') = k+2+d-1 = M$. Since $z \neq 0$, it follows that $L(t^{k+2}p^2) = 0$. The polynomial $t^{k+2}p$ is monic of degree $M+1$, which extends the orthogonality by one degree. Induction shows that $L(pq) = 0$ for every polynomial q.

Approximate the continuous function $\overline{p(t)}e^{-\mathrm{i}zt}$ uniformly on $[-1, 1]$ by complex polynomials. Passage to the limit gives

$$\int_{-1}^{1} |p(t)|^2 (1-t^2)^\alpha \, \mathrm{d}t = 0,$$

contradicting the fact that p is monic. □

Remark 2.2. The same proof applies to the nonsymmetric Jacobi weight $(1-t)^a(1+t)^b e^{\mathrm{i}zt}$, $a, b > -1$. Its Pearson bracket is $\mathrm{i}z(1-t^2) + (b-a) - (a+b+2)t$. The extra constant term does not change the degree induction. The positivity theorem uses symmetry and is asserted only for $a = b$.

Because $(\mu_j^{(\alpha)})' = \mathrm{i}\mu_{j+1}^{(\alpha)}$, the Desnanot–Jacobi identity for consecutive derivative Hankel determinants gives

$$D_N D_N'' - (D_N')^2 = -D_{N-1} D_{N+1}, \qquad N \geq 1. \tag{2.3}$$

Here and below frequency derivatives are taken with α fixed. The sign can already be read at $N = 1$: $\mu_0\mu_0'' - (\mu_0')^2 = -\mu_0\mu_2 + \mu_1^2 = -D_2$.

Corollary 2.3. *Every complex zero of every $D_N^{(\alpha)}$, $N \geq 1$, is simple.*

Proof. If $D_N = D_N' = 0$ at a nonzero frequency, (2.3) forces one of the adjacent determinants to vanish, contrary to Lemma 2.1. Frequency zero is not a zero, because its moment matrix is positive definite. □

For later normalisation, the zero-frequency value is

$$D_N^{(\alpha)}(0) = 2^{N(2\alpha+N)} \prod_{j=0}^{N-1} \frac{j!\,\Gamma(\alpha+j+1)^2}{\Gamma(2\alpha+N+j+1)} > 0. \tag{2.4}$$

This is the Selberg integral after $t = 2s-1$, or equivalently the product of the monic Jacobi polynomial norms. Positivity alone in the preceding argument does not require this evaluation.

3 Endpoint expansions, uniform in the exponent

Let

$$L_m(\alpha) = \prod_{j=0}^{m-1} j!\,\Gamma(\alpha+j+1), \qquad L_0 = 1. \tag{3.1}$$

Andréief's identity yields

$$D_N^{(\alpha)}(x) = \frac{1}{N!}\int_{[-1,1]^N} \Delta(t)^2 \prod_{j=1}^{N}(1-t_j^2)^{\alpha} e^{\mathrm{i}x\sum_j t_j}\,\mathrm{d}t_1\cdots\mathrm{d}t_N, \tag{3.2}$$

where $\Delta(t) = \prod_{i<j}(t_j - t_i)$.

The whole large-frequency analysis rests on one elementary fact about one-variable endpoint integrals, which we prove first with constants that are uniform in the exponent. Throughout, $K \subset (-1,\infty)$ is a compact interval, and $O_K(\cdot)$ denotes a bound whose constant depends on K and on the displayed fixed integers, but not on $\alpha \in K$ or on x.

Lemma 3.1 (uniform endpoint lemma)**.** *Fix $\delta \in (0,1]$, a smooth cutoff χ on $[0,\infty)$ with $\chi = 1$ on $[0,\delta/3]$ and $\chi = 0$ on $[2\delta/3,\infty)$, an integer $q \ge 0$, and a function $h(s,\alpha)$, smooth in $s \in [0,\delta]$, whose s-derivatives of every order are bounded and jointly continuous on $[0,\delta]\times K$. Then for every integer $M \ge 1$ and $x \ge 1$,*

$$\int_0^\delta s^{\alpha+q} h(s,\alpha)\chi(s) e^{-\mathrm{i}xs}\,\mathrm{d}s = \sum_{m=0}^{M-1} \frac{\partial_s^m h(0,\alpha)}{m!}\,\Gamma(\alpha+q+m+1)(\mathrm{i}x)^{-(\alpha+q+m+1)} + O_K(x^{-(q+M)}), \tag{3.3}$$

with the principal branch of $(\mathrm{i}x)^{-c} = x^{-c}e^{-\mathrm{i}\pi c/2}$. The coefficients are continuous in $\alpha \in K$. The same holds with $e^{+\mathrm{i}xs}$ and $(-\mathrm{i}x)$ in place of $e^{-\mathrm{i}xs}$ and $(\mathrm{i}x)$.

Proof. Taylor's theorem gives $h(s,\alpha) = \sum_{m<M} h_m(\alpha)s^m + s^M R(s,\alpha)$ with $h_m = \partial_s^m h(0,\alpha)/m!$, where every s-derivative of R is bounded on $[0,\delta]\times K$. Put $n = q+M$ and $g(s) = s^{\alpha+q+M}R(s,\alpha)\chi(s)$. For $k \le n$ the derivative $g^{(k)}$ is a finite sum of terms bounded by a constant times $s^{\alpha+q+M-k}$, and $\alpha+q+M-k \ge \alpha > -1$. Hence every $g^{(k)}$, $k \le n$, is integrable on $[0,\delta]$ with $\int_0^\delta |g^{(k)}| \le C_K$, since $\int_0^\delta s^\alpha\,\mathrm{d}s \le \delta^{\alpha+1}/(\alpha+1)$ is bounded on K. For $k < n$, $g^{(k)}(s) = O(s^{\alpha+1}) \to 0$ as $s \to 0$, and $g^{(k)}(\delta) = 0$. Integrating by parts n times therefore gives $\bigl|\int_0^\delta g\,e^{-\mathrm{i}xs}\,\mathrm{d}s\bigr| \le C_K x^{-n}$.

It remains to treat $\int_0^\infty s^\beta\chi(s)e^{-\mathrm{i}xs}\,\mathrm{d}s$ with $\beta = \alpha+q+m$. Insert the factor $e^{-\varepsilon s}$, $\varepsilon > 0$. Then

$$\int_0^\infty s^\beta e^{-(\varepsilon+\mathrm{i}x)s}\,\mathrm{d}s = \Gamma(\beta+1)(\varepsilon+\mathrm{i}x)^{-\beta-1}$$

and the difference from the cutoff integral is $\int_0^\infty s^\beta(1-\chi(s))e^{-(\varepsilon+\mathrm{i}x)s}\,\mathrm{d}s$. The factor $1-\chi$ vanishes near zero. Choose one integer $L \ge q+M$ with $L > \max_{\alpha\in K,\,m<M}(\alpha+q+m+1)$, and integrate by parts L times; for $\varepsilon > 0$ all boundary terms vanish. By the Leibniz rule the Lth derivative of $s^\beta(1-\chi(s))e^{-\varepsilon s}$ is a sum of compactly supported cutoff-derivative terms and terms $\varepsilon^j s^{\beta-L+j}e^{-\varepsilon s}$ on $s \ge \delta/3$, and $\varepsilon^j s^j e^{-\varepsilon s} \le C_j$. Hence it is dominated, uniformly in $\varepsilon \in (0,1]$ and $\alpha \in K$, by an integrable multiple of $s^{\beta-L}$ on $[\delta/3,\infty)$. This difference is therefore $O_K(x^{-L})$ uniformly in ε. Letting $\varepsilon \to 0$ by dominated convergence on the cutoff side gives $\int_0^\infty s^\beta\chi e^{-\mathrm{i}xs}\,\mathrm{d}s = \Gamma(\beta+1)(\mathrm{i}x)^{-\beta-1} + O_K(x^{-L})$, with L as large as required. Summing over $m < M$ proves (3.3). Continuity of the coefficients follows from continuity of Γ and of h_m. When $\alpha > 0$ the remainder bound is weaker than the last retained term; this is harmless because M may be taken as large as required. □

Lemma 3.2 (moment splitting). *Fix $\delta \in (0, 1/2]$ and smooth functions χ_+, χ_0, χ_- on $[-1, 1]$, with values in $[0, 1]$ and sum one, where $\chi_+(t) = \chi(1-t)$, $\chi_-(t) = \chi(1+t)$, and χ_0 is supported in $[-1+\delta/3, 1-\delta/3]$. For each $j \geq 0$ write $\mu_j^{(\alpha)} = \mu_j^+ + \mu_j^0 + \mu_j^-$ according to this partition. Then, for real $x \geq 1$, $\mu_j^0(x) = O_K(x^{-L})$ for every L, while $e^{-\mathrm{i}x}\mu_j^+(x)$ and $e^{\mathrm{i}x}\mu_j^-(x)$ have expansions of the form (3.3) to every order, with coefficients continuous in $\alpha \in K$.*

Proof. On the support of χ_0 the function $t^j(1-t^2)^\alpha \chi_0(t)$ is smooth, compactly supported in $(-1, 1)$, and has all derivatives bounded on K; repeated integration by parts gives the first claim. Near $t = 1$, put $t = 1 - s$. Then $t^j(1-t^2)^\alpha e^{\mathrm{i}xt} = e^{\mathrm{i}x} s^\alpha\,(1-s)^j(2-s)^\alpha e^{-\mathrm{i}xs}$, and $h(s, \alpha) = (1-s)^j(2-s)^\alpha$ satisfies the hypotheses of Lemma 3.1 on $[0, \delta]$, because $2 - s \geq 3/2$ there. The endpoint $t = -1$ is symmetric. □

Proposition 3.3. *Fix $N \geq 1$ and a compact interval $K \subset (-1, \infty)$. For real $x \geq 1$, $D_N^{(\alpha)}(x)$ is a sum over endpoint allocations $k + l = N$ of terms $e^{\mathrm{i}(k-l)x} F_{k,l}(x, \alpha)$ plus $O_K(x^{-L})$ for any prescribed L. Each $F_{k,l}$ has an asymptotic expansion in powers $x^{-N(\alpha+1)-m}$, $m = 0, 1, 2, \ldots$, to every order, with coefficients continuous in $\alpha \in K$ and remainders uniform for $\alpha \in K$. The first nonzero term of $e^{\mathrm{i}(k-l)x} F_{k,l}$ is*

$$2^{N\alpha+2kl} L_k(\alpha) L_l(\alpha) x^{-E_k} \exp\left(\mathrm{i}(k-l)\left[x - \frac{\pi}{2}(\alpha + N)\right]\right), \quad E_k = N\alpha + k^2 + l^2. \tag{3.4}$$

The frequency derivative $\partial_x D_N^{(\alpha)}$ has the termwise-differentiated expansion, with the same uniformity.

Proof. Reduction to one-variable integrals. Expanding $\Delta(t)^2$ into monomials in (3.2) expresses $D_N^{(\alpha)}$ as a fixed finite linear combination of products $\prod_{i=1}^N \mu_{a_i}^{(\alpha)}$ (this is just the Leibniz expansion of the Hankel determinant). Insert the splitting of Lemma 3.2 into each factor. Any product containing a factor μ^0 is $O_K(x^{-L})$, since every $|\mu_j^\pm|$ is bounded by $\int_{-1}^1 (1-t^2)^\alpha \,\mathrm{d}t$, which is bounded on K. Collect the remaining products according to how many factors carry $+$ and how many carry $-$. Equivalently, for a fixed allocation in which k coordinates carry χ_+ and l carry χ_-, the corresponding term of (3.2) is

$$\binom{N}{k} \frac{1}{N!} \int \Delta(t)^2 \prod_{i \leq k} (1-t_i^2)^\alpha \chi_+(t_i) e^{\mathrm{i}xt_i} \prod_{i > k} (1-t_i^2)^\alpha \chi_-(t_i) e^{\mathrm{i}xt_i} \,\mathrm{d}t.$$

Put $t_i = 1 - s_i$ for $i \leq k$ and $t_i = -1 + s_i'$ for $i > k$. The square $\Delta(t)^2$ becomes the *polynomial* $\Delta(s)^2 \Delta(s')^2 \prod_{i \leq k < j} (2 - s_i - s_j')^2$. All other factors are products of one-variable functions of the type in Lemma 3.2. Expanding this polynomial into monomials therefore writes the allocation term exactly as $e^{\mathrm{i}(k-l)x} F_{k,l}$, where $F_{k,l}$ is a finite sum of products of one-variable integrals covered by Lemma 3.1. No multivariable Taylor remainder occurs, and the uniformity and continuity statements follow by multiplying finitely many uniform one-variable expansions.

The leading term. A monomial of total degree d in the s, s' variables contributes, by (3.3), terms of order $x^{-N(\alpha+1)-d-m}$, $m \geq 0$. The polynomial $\Delta(s)^2 \Delta(s')^2$ is homogeneous of degree $k(k-1) + l(l-1)$ and is the lowest-degree part of the full polynomial; its coefficient multiplies 2^{2kl} from the cross factor at $s = s' = 0$. The constant terms of the smooth factors are $h(0, \alpha) = 2^\alpha$ per coordinate, giving $2^{N\alpha}$. The order is therefore $N(\alpha+1) + k(k-1) + l(l-1) = E_k$, and the leading coefficient is obtained by replacing each one-variable integral by its first term in (3.3). That replacement is the same as evaluating

$$\int_{[0,\infty)^m} \Delta(u)^2 \prod_{j=1}^m u_j^\alpha e^{-s\sum_j u_j} \,\mathrm{d}u = m! L_m(\alpha) s^{-m(\alpha+m)}, \qquad \Re s > 0, \tag{3.5}$$

at the boundary values $s = \pm \mathrm{i}x$, where the right-hand side is read as the analytic continuation from $\Re s > 0$. Indeed, both sides of (3.5) are sums of the same products of Gamma values times powers of s, and (3.3) supplies exactly these products with $s = \mathrm{i}x$ (respectively $-\mathrm{i}x$). At $s = 1$, Andréief reduces the integral to $m! \det[\Gamma(\alpha + i + j + 1)]$. Factoring $\Gamma(\alpha + i + 1)$ from row i leaves $(\alpha + i + 1)_j$, a monic polynomial of degree j in i. Its determinant is the Vandermonde determinant at $0, \dots, m-1$, namely $\prod_{j=0}^{m-1} j!$. Homogeneity gives the formula for general s. The boundary values supply the phases $e^{\mp \mathrm{i}\pi m(\alpha+m)/2}$. The factor $\binom{N}{k}$ and $1/N!$ cancel the two factorials in (3.5). The phase simplifies using $k(\alpha + k) - l(\alpha + l) = (k - l)(\alpha + N)$, giving (3.4). This coefficient is nonzero, so it is the first nonzero term.

The derivative. For the one-variable integrals $I_q(x) = \int s^{\alpha+q} h\chi e^{-\mathrm{i}xs}\, \mathrm{d}s$, differentiation under the integral gives $I_q' = -\mathrm{i}I_{q+1}$. Applying Lemma 3.1 to both sides and using $\Gamma(p+1) = p\Gamma(p)$ shows that the expansion of I_q' is the termwise derivative of that of I_q; the conjugate phase is analogous. The product rule, applied to each allocation term including its phase $e^{\mathrm{i}(k-l)x}$, then gives the uniform expansion of $\partial_x D_N^{(\alpha)}$ as the termwise derivative, without differentiating an uncontrolled remainder.

Joint continuity. By dominated convergence, every $\mu_j^{(\alpha)}(x)$ and its x-derivatives are jointly continuous in $(\alpha, x) \in (-1, \infty) \times \mathbb{R}$; hence so are $D_N^{(\alpha)}(x)$ and $\partial_x D_N^{(\alpha)}(x)$. This is used in Section 4. □

Corollary 3.4. *For $r \geq 1$, uniformly on compact $K \subset (-1, \infty)$,*

$$D_{2r}^{(\alpha)}(x) = C_{r,\alpha} x^{-2r(\alpha+r)}(1 + O_K(x^{-2})), \qquad C_{r,\alpha} = 2^{2r(\alpha+r)} L_r(\alpha)^2 > 0. \tag{3.6}$$

In particular, there is a common R such that $D_{2r}^{(\alpha)}(x) > 0$ for every $\alpha \in K$ and $x \geq R$.

Proof. Write $k = r + s$, $l = r - s$ in (3.4). Then $E_k = 2r(\alpha + r) + 2s^2$. The balanced allocation $k = l = r$ is the unique leading term, and its phase is one. Its first correction cancels between the conjugate endpoint groups; the explicit calculation below also verifies this cancellation. Every unbalanced allocation is smaller by at least x^{-2}. The coefficient is continuous and strictly positive on K, so the uniform expansion implies a common positivity threshold. □

The odd case needs a first correction for the degree calculations later.

Proposition 3.5. *Let $r \geq 0$, set $N = 2r + 1$, and put*

$$E = (2r+1)\alpha + 2r^2 + 2r + 1, \qquad \theta = x - \frac{\pi\alpha}{2}, \tag{3.7}$$

$$A_{r,\alpha} = 2^{(2r+1)\alpha + 2r(r+1)+1} L_r(\alpha) L_{r+1}(\alpha), \qquad T_{r,\alpha} = \frac{\alpha(\alpha + 2r + 1)}{2} + r(r+1). \tag{3.8}$$

Then, uniformly on compact exponent intervals,

$$D_{2r+1}^{(\alpha)}(x) = (-1)^r A_{r,\alpha} x^{-E} \left[\sin\theta + \frac{T_{r,\alpha}}{x} \cos\theta + O_K(x^{-2}) \right]. \tag{3.9}$$

The bracket remainder can be differentiated once with error $O_K(x^{-2})$.

Proof. The leading allocations are $(k, l) = (r+1, r)$ and $(r, r+1)$. Their phases in (3.4) are conjugate and add to $2(-1)^r \sin\theta$. The next allocations are down by x^{-4}. In the scaled endpoint variables, the linear correction to the smooth amplitude at an allocation (k, l) is

$$-\frac{1}{x}\left[\left(\frac{\alpha}{2} + l\right) \sum_{j=1}^{k} u_j + \left(\frac{\alpha}{2} + k\right) \sum_{j=1}^{l} v_j \right]. \tag{3.10}$$

Differentiate (3.5) in s. For the group with $e^{-\mathrm{i}\sum u_j}$, the normalised endpoint moment of $\sum u_j$ is $-\mathrm{i}k(\alpha+k)$; the conjugate group has moment $\mathrm{i}l(\alpha+l)$. For $(k,l)=(r+1,r)$, the relative correction is $\mathrm{i}T_{r,\alpha}/x$, where

$$T_{r,\alpha}=\left(\frac{\alpha}{2}+r\right)(r+1)(\alpha+r+1)-\left(\frac{\alpha}{2}+r+1\right)r(\alpha+r).$$

Expansion gives (3.8). Adding the conjugate allocation proves (3.9). The remainder and its differentiated version follow from Proposition 3.3. For the balanced even allocation, the two terms in (3.10) instead cancel, as used in Corollary 3.4. □

4 Continuation and the real zeros

Proof of Theorem 1.1. For real frequency, reflection of the integration variable gives $\overline{\mu_j^{(\alpha)}(x)}=(-1)^j\mu_j^{(\alpha)}(x)$. The row and column sign exponents of a size-N determinant sum to $N(N-1)$, an even number. Thus $D_N^{(\alpha)}(x)$ is real; the same calculation shows it is even in x.

Fix $2r$ and a target exponent $\alpha_*>-1$. Let K be the closed interval joining 0 and α_*. Positive definiteness at frequency zero, compactness, and continuity give $\varepsilon>0$ such that $D_{2r}^{(\alpha)}(x)>0$ for all $\alpha\in K$ and $|x|\le\varepsilon$. Corollary 3.4 gives $R>\varepsilon$ with the same positivity for $|x|\ge R$.

Define

$$S=\{\alpha\in K: D_{2r}^{(\alpha)}(x)>0 \text{ for every } x\in[\varepsilon,R]\}.$$

The exponent-zero theorem [1, Theorem 4.3] states that for every $n\ge 0$ and every real frequency the monic kissing polynomial of degree $2n$ for $e^{\mathrm{i}\omega t}$ on $[-1,1]$ exists; it is unconditional, with no asymptotic restriction. It gives $0\in S$. Indeed, it proves nonvanishing, and positivity follows by continuity from $x=0$. The set S is relatively open, since a positive continuous function on this compact frequency interval has a positive minimum.

It is also relatively closed. If $\alpha_j\in S$ tends to $\alpha\in K$, then $D_{2r}^{(\alpha)}\ge 0$ on $[\varepsilon,R]$. A zero would lie strictly inside the interval because its endpoints are already in positive regions. At such an interior minimum the frequency derivative would also vanish, contradicting Corollary 2.3. Hence $\alpha\in S$. Connectedness gives $S=K$, and evenness completes the proof for every real frequency. □

Corollary 4.1. *For every real $\alpha>-1$ and every $r\ge 0$, the function $D_{2r+1}^{(\alpha)}$ has infinitely many positive real zeros, all simple. For all sufficiently large integers m, exactly one zero lies near $x_m=m\pi+\pi\alpha/2$, and it equals*

$$x_m-\frac{T_{r,\alpha}}{x_m}+O(m^{-2}). \tag{4.1}$$

This local label need not be the global index of that zero.

Proof. The leading sine in (3.9) has alternating signs at its successive extrema. The differentiated expansion gives a derivative of fixed nonzero sign in a small neighbourhood of each sufficiently large phase zero, and the function is bounded away from zero outside these neighbourhoods. This proves existence and uniqueness there. Substituting $x=x_m+\delta$ in (3.9) gives $\delta=-T_{r,\alpha}/x_m+O(m^{-2})$. Simplicity of every zero, including the remaining ones, is Corollary 2.3. □

The Toda identity and Theorem 1.1 also give a direct consequence between successive odd-size zeros:

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}\log|D_{2r+1}^{(\alpha)}(x)| = -\frac{D_{2r}^{(\alpha)}(x)D_{2r+2}^{(\alpha)}(x)}{D_{2r+1}^{(\alpha)}(x)^2} < 0. \tag{4.2}$$

This strict logarithmic concavity does not by itself locate the first positive zero.

5 The degree of the Hankel polynomials

Let u be a polynomial variable. For a parameter λ, set $P_{-1} = 0$, $P_0 = 1$, $P_1 = -2\lambda$, and

$$P_{k+2} = -(k + 2\lambda + 1)P_{k+1} - uP_k - kuP_{k-1}, \qquad k \geq 0. \tag{5.1}$$

Define

$$H_n(u;\lambda) = \det[P_{i+j}(u;\lambda)]_{i,j=0}^{n-1}, \qquad H_0 = 1. \tag{5.2}$$

These are precisely the polynomials in [2, pp. 50–53], with $u = \zeta^2$. Put

$$a = \lambda + \frac{1}{2}, \qquad K_r(a) = \prod_{j=0}^{r-1} j!(a)_j, \qquad K_0 = 1. \tag{5.3}$$

Theorem 5.1. *For every real $\lambda > -\frac{1}{2}$ and every $n \geq 0$,*

$$\deg_u H_n = \left\lfloor \frac{n^2}{4} \right\rfloor. \tag{5.4}$$

The leading coefficients $\ell_n(\lambda)$ are

$$\ell_{2r}(\lambda) = (-1)^r 2^{2r(r-1)} K_r(a)^2, \qquad r \geq 1, \tag{5.5}$$

$$\ell_{2r+1}(\lambda) = 2^{2r^2} K_r(a)K_{r+1}(a), \qquad r \geq 0. \tag{5.6}$$

The degree upper bound and these coefficient formulas hold as polynomial identities for every complex λ. Equality in the degree holds whenever the displayed coefficient is nonzero.

Proof. First assume $\lambda > -\frac{1}{2}$ is real and set $\alpha = \lambda - \frac{1}{2}$. The Fourier–Bessel integral is

$$\mu_0^{(\alpha)}(\zeta) = \sqrt{\pi}\,\Gamma\left(\lambda + \frac{1}{2}\right)\left(\frac{2}{\zeta}\right)^{\lambda} J_\lambda(\zeta). \tag{5.7}$$

Let ζ run through an unbounded sequence of positive zeros of $J_{\lambda-1}$. At these frequencies the Bessel recurrence gives $\mu_1 = 2\mathrm{i}\lambda\mu_0/\zeta$. The Pearson identity for a monomial gives

$$\mathrm{i}\zeta\mu_{k+2} = k\mu_{k-1} - (k + 2\lambda + 1)\mu_{k+1} + \mathrm{i}\zeta\mu_k.$$

Comparison with (5.1) proves

$$\mu_j^{(\alpha)}(\zeta) = \mu_0^{(\alpha)}(\zeta)(\mathrm{i}\zeta)^{-j} P_j(\zeta^2;\lambda). \tag{5.8}$$

Factoring rows and columns gives the exact identity

$$D_n^{(\alpha)}(\zeta) = \mu_0^{(\alpha)}(\zeta)^n(\mathrm{i}\zeta)^{-n(n-1)} H_n(\zeta^2;\lambda). \tag{5.9}$$

This step assumes neither existence of an orthogonal-polynomial sequence nor rationality of λ.

Write $\theta = \zeta - \pi\alpha/2$ and $B = 2^{\alpha+1}\Gamma(\alpha+1)$. The large-argument Bessel expansion at zeros of $J_{\lambda-1}$ gives

$$\cos\theta = O(\zeta^{-1}), \qquad \sin^2\theta = 1 + O(\zeta^{-2}), \qquad \mu_0^{(\alpha)}(\zeta) = B\zeta^{-\alpha-1}(\sin\theta + O(\zeta^{-1})). \tag{5.10}$$

In particular μ_0 is nonzero along all sufficiently large members of the sequence.

For $n = 2r$, insert (3.6) and (5.10) into (5.9). Since $\mathrm{i}^{n(n-1)} = (-1)^r$ and $(\sin\theta)^{2r} \to 1$,

$$\frac{H_{2r}(\zeta^2;\lambda)}{\zeta^{2r^2}} \longrightarrow (-1)^r \frac{2^{2r(\alpha+r)} L_r(\alpha)^2}{B^{2r}} = (-1)^r 2^{2r(r-1)} K_r(a)^2.$$

For $n = 2r+1$, the leading odd term is $(-1)^r A_{r,\alpha}\zeta^{-E}\sin\theta$. Again $\mathrm{i}^{n(n-1)} = (-1)^r$, so the two signs cancel and

$$\frac{H_{2r+1}(\zeta^2;\lambda)}{\zeta^{2r(r+1)}} \longrightarrow \frac{A_{r,\alpha}}{B^{2r+1}} = 2^{2r^2} K_r(a) K_{r+1}(a).$$

Each limit is finite and nonzero because $a > 0$. Since H_n is a polynomial in u, its growth along the unbounded sequence $u = \zeta^2$ determines its degree and leading coefficient: a higher degree would give an unbounded normalised sequence, and a lower degree would give limit zero.

Finally, every coefficient of H_n is a polynomial in λ. The coefficients above the stated degree vanish on the real interval $(-\frac{1}{2}, \infty)$ and therefore vanish identically. The leading coefficient identities extend in the same way. □

Rearranging the products in (5.5)–(5.6) gives the equivalent formulas

$$\ell_{2r} = (-1)^r 2^{r(r-1)} \prod_{j=1}^{r-1} [j(2\lambda + 2j - 1)]^{2(r-j)}, \tag{5.11}$$

$$\ell_{2r+1} = 2^{r^2} \prod_{j=1}^{r} [j(2\lambda + 2j - 1)]^{2(r-j)+1}. \tag{5.12}$$

These are the products proposed in [4, Remark 2.6]. The degree statement is the first conjecture on p. 53 of [2].

6 The shifted-numerator degree

For $n \geq 1$, define the shifted-last-column determinant

$$\widetilde{H}_n = \det[P_{i+j+\mathbf{1}_{\{j=n-1\}}}]_{i,j=0}^{n-1}, \qquad V_n = \widetilde{H}_n H_{n+1} - \widetilde{H}_{n+1} H_n. \tag{6.1}$$

Thus the last column uses $P_n, \ldots, P_{2n-1}$; all earlier columns are unchanged. The prime used for this expression in [2] denotes a shifted determinant, not a parameter derivative.

Theorem 6.1. *As a formal polynomial over $\mathbb{Q}(\lambda)$,*

$$\deg_u V_n = \frac{n(n+1)}{2}. \tag{6.2}$$

The coefficient of $u^{n(n+1)/2}$ is

$$C_n(\lambda)\ell_n(\lambda)\ell_{n+1}(\lambda), \tag{6.3}$$

where

$$C_{2r}(\lambda) = (r+2)\lambda + \frac{r(2r+5)}{2}, \qquad r \geq 1, \tag{6.4}$$

$$C_{2r+1}(\lambda) = 1 + \frac{3r}{2} - r^2 - r\lambda, \qquad r \geq 0. \tag{6.5}$$

For real $\lambda > -\frac{1}{2}$, the specialised degree equals $n(n+1)/2$ except at $(n,\lambda) = (3,3/2)$ and $(5,0)$, where it is respectively 5 and 14.

Proof. Define $\widetilde{D}_n$ by the same shifted-last-column construction with the moments $\mu_j^{(\alpha)}$. Differentiate D_n by columns. Each column derivative other than the last duplicates the next column, so

$$D_n'(x) = \mathrm{i}\widetilde{D}_n(x). \tag{6.6}$$

This holds at every frequency, before any specialisation. At a positive zero ζ of $J_{\lambda-1}$, (5.8) gives

$$\widetilde{D}_n(\zeta) = \mu_0(\zeta)^n(\mathrm{i}\zeta)^{-n(n-1)-1}\widetilde{H}_n(\zeta^2;\lambda).$$

Consequently,

$$\frac{V_n}{H_nH_{n+1}} = \zeta\left(\frac{D_n'(\zeta)}{D_n(\zeta)} - \frac{D_{n+1}'(\zeta)}{D_{n+1}(\zeta)}\right). \tag{6.7}$$

No identity valid only on the discrete Bessel-zero set has been differentiated.

Put $\alpha = \lambda - \frac{1}{2}$. The even expansion and its derivative give

$$\lim_{x\to\infty} x\frac{(D_{2r}^{(\alpha)})'(x)}{D_{2r}^{(\alpha)}(x)} = -2r(\alpha+r). \tag{6.8}$$

For the odd limit the first correction in Proposition 3.5 matters. The first two Bessel terms are

$$J_{\alpha-1/2}(x) = \sqrt{\frac{2}{\pi x}}\left[\cos\theta - \frac{B_\alpha}{x}\sin\theta + O(x^{-2})\right], \qquad B_\alpha = \frac{\alpha(\alpha-1)}{2}.$$

Thus, at its large positive zeros, $\zeta\cot\theta \to B_\alpha$. Taking the logarithmic derivative of (3.9) yields

$$\begin{aligned}\lim_{\zeta\to\infty}\zeta\frac{(D_{2r+1}^{(\alpha)})'(\zeta)}{D_{2r+1}^{(\alpha)}(\zeta)} &= -E + B_\alpha - T_{r,\alpha}\\ &= -[(3r+2)\alpha + 3r^2 + 3r + 1].\end{aligned} \tag{6.9}$$

Here $B_\alpha - T_{r,\alpha} = -(r+1)(\alpha+r)$. The relevant determinants are nonzero along all sufficiently large members of this sequence by the same endpoint estimates.

Subtracting (6.8) and (6.9) in the appropriate order shows that the limit in (6.7) is $C_n(\lambda)$. Since

$$\deg H_n + \deg H_{n+1} = \left\lfloor\frac{n^2}{4}\right\rfloor + \left\lfloor\frac{(n+1)^2}{4}\right\rfloor = \frac{n(n+1)}{2},$$

boundedness of the ratio along an unbounded positive sequence proves the degree upper bound, and its limit gives (6.3). The coefficients are polynomials in λ, so these statements extend identically from $\lambda > -\frac{1}{2}$. Since (6.3) is not identically zero, the formal degree is exact.

For $\lambda > -\frac{1}{2}$, every ℓ_n is nonzero. The factor C_{2r} is positive. The factor C_1 is one, and for $r \geq 1$ the unique zero of C_{2r+1} is $\lambda = \frac{3}{2} - r + 1/r$. Only $r = 1, 2$ put this zero in the admissible parameter interval. The exact calculations below determine the remaining specialised degrees. □

At $\lambda = 3/2$, the defining recurrence gives

$$V_3 = -4608(13u^5 - 402u^4 + 4428u^3 - 18441u^2 + 14175u - 3645). \tag{6.10}$$

At $\lambda = 0$, it gives

$$\begin{aligned} V_5 = 181440u^9(208u^5 - 9468u^4 + 154323u^3 \\ - 973368u^2 + 1369008u - 648000). \end{aligned} \tag{6.11}$$

Their leading coefficients are -59904 and 37739520. These are finite exact polynomial identities, not numerical evaluations of Bessel functions. They can be checked directly from (5.1), (5.2) and (6.1).

The source writes its coefficients as functions of λ, so the natural reading of its second degree conjecture is the formal generic statement. Theorem 6.1 confirms that statement and refines it at exceptional specialisations; the exceptions are not presented as a refutation of that generic interpretation.

7 The recurrence-coefficient conjectures

Use the recurrence convention

$$\pi_{n+1}(t) = (t - \mathrm{i}\alpha_n)\pi_n(t) - \beta_n\pi_{n-1}(t).$$

At the Bessel-zero frequencies under consideration, its determinant formulas are

$$\alpha_n = \frac{1}{\zeta}\left(\frac{\widetilde{H}_n}{H_n} - \frac{\widetilde{H}_{n+1}}{H_{n+1}}\right), \qquad \beta_n = -\frac{H_{n-1}H_{n+1}}{uH_n^2}, \qquad u = \zeta^2. \tag{7.1}$$

They hold as rational-function identities, and give actual recurrence coefficients whenever the required determinants are nonzero. In the source regime of rational $\lambda > -\frac{1}{2}$ and nonzero zeros of $J_{\lambda-1}$, nonvanishing follows from [2, Theorem 2.2].

7.1 The general parameter

Put $d_n = \lfloor n^2/4 \rfloor$, $E_n = H_n/\ell_n$, and $E_0 = E_1 = 1$. The limits (6.8)–(6.9) show that $\deg \widetilde{H}_n \le d_n$ and that $\widetilde{H}_n/H_n \to b_n$, where

$$b_{2r} = -2r(\lambda - \tfrac{1}{2} + r), \qquad b_{2r+1} = -(3r+2)\lambda - 3r^2 - \tfrac{3}{2}r.$$

Set $S_n = n(2\lambda + n - 1)$ and $e_n = S_n + b_n$. Thus

$$e_{2r} = r(2\lambda + 2r - 1), \qquad e_{2r+1} = \frac{r}{2}(2\lambda + 2r + 1). \tag{7.2}$$

For $n \ge 2$, these are positive when $\lambda > -\frac{1}{2}$.

Lemma 7.1. *The polynomial identity $\widetilde{H}_n(0) + S_nH_n(0) = 0$ holds for every $n \ge 1$. Consequently*

$$F_n(u) = \frac{\widetilde{H}_n(u) + S_nH_n(u)}{u\ell_ne_n}, \qquad n \ge 2, \tag{7.3}$$

is monic of degree $d_n - 1$.

Proof. At $u = 0$, (5.1) gives $P_j(0) = (-1)^j (b)_j$, $b = 2\lambda$. Factor $(b)_i$ from row i and remove the row and column signs. The unshifted determinant is then the evaluation determinant of the monic polynomials $q_j(x) = (b + x)_j$, $0 \le j < n$, at $x = 0, \dots, n-1$. Its determinant is the corresponding Vandermonde. The shifted last column uses q_n. Its coefficient of x^{n-1} is $nb + n(n-1)/2$. Reducing it modulo $\prod_{i=0}^{n-1}(x - i)$ adds a further $n(n-1)/2$ to that coefficient. The ratio of the two evaluation determinants is therefore $n(b + n - 1) = S_n$. The extra column shift changes the original sign by minus one. This proves the identity whenever the factored quantities are nonzero, hence identically in λ.

The numerator in (7.3) is divisible by u. Its leading coefficient is $\ell_n(S_n + b_n) = \ell_n e_n \neq 0$, proving the degree and monicity assertion. □

Define $X_{d_n} = E_n$ for $n \ge 1$ and $X_{d_n - 1} = F_n$ for $n \ge 4$. These definitions are consistent: the sequence d_n, $n \ge 1$, is strictly increasing, and $d_{n-1} < d_n - 1 < d_n$ for $n \ge 4$. There are therefore no collisions between numerator and denominator labels. In particular $X_0 = 1$, $X_1 = u - 2\lambda$, and every assigned X_j is monic of degree j; unused labels may be filled arbitrarily.

Theorem 7.2. *All four formulas of [4, Conjecture 2.5] hold in their stated ranges. Explicitly, with every X evaluated at $u = \zeta^2$,*

$$\alpha_{2k} = \frac{2(\lambda + 2k)}{\zeta} + k\zeta \left[(2\lambda + 2k - 1) \frac{X_{k^2-1}}{X_{k^2}} - \frac{2\lambda + 2k + 1}{2} \frac{X_{k(k+1)-1}}{X_{k(k+1)}} \right], \qquad k \ge 2, \tag{7.4}$$

$$\alpha_{2k+1} = \frac{2(\lambda + 2k + 1)}{\zeta} + (2\lambda + 2k + 1)\zeta \left[\frac{k}{2} \frac{X_{k(k+1)-1}}{X_{k(k+1)}} - (k+1) \frac{X_{k(k+2)}}{X_{(k+1)^2}} \right], \qquad k \ge 2, \tag{7.5}$$

$$\beta_{2k} = -2k(2\lambda + 2k - 1) \frac{X_{k(k-1)} X_{k(k+1)}}{u X_{k^2}^2}, \qquad k \ge 1, \tag{7.6}$$

$$\beta_{2k+1} = \frac{X_{k^2} X_{(k+1)^2}}{u X_{k(k+1)}^2}, \qquad k \ge 0. \tag{7.7}$$

Proof. By (7.3), $\widetilde{H}_n / H_n = -S_n + u e_n F_n / E_n$. Substitution in (7.1), using $S_{n+1} - S_n = 2(\lambda + n)$, gives

$$\alpha_n = \frac{2(\lambda + n)}{\zeta} + \zeta \left(e_n \frac{F_n}{E_n} - e_{n+1} \frac{F_{n+1}}{E_{n+1}} \right).$$

The values (7.2) give the first two formulas. The ratios of the leading coefficients (5.5)–(5.6) give the last two. □

The lower-index restriction on the α formulas is necessary for this common-index notation. Directly, $F_2 = 1$, $F_3 = u - 4\lambda^2 - 4\lambda$, and $E_2 = u - 2\lambda$. The last two degree-one polynomials are generally different. The source uses the displayed common-index formulas only for $k \ge 2$; its separate formulas for α_2, α_3 retain the appropriate small-index numerator. For fixed admissible λ the constructed X_j are ordinary polynomials in u; formally their coefficients lie in $\mathbb{Q}(\lambda)$.

7.2 The exceptional parameter $\lambda = 0$

At $\lambda = 0$, powers of u cancel from the Hankel determinants, leading to the different index pattern of [4, Conjecture 2.7]. We now establish that pattern and all its recurrence formulas.

Lemma 7.3. *For $n \geq 1$, the exact valuation of $H_n(u;0)$ at $u = 0$ is $n-1$. Writing $m = n-1$, its first coefficient is*

$$[u^{n-1}]H_n(u;0) = \left(-\frac{1}{2}\right)^m \left(1 + \frac{m(m+1)}{2}\right) \prod_{j=0}^{m-1} j!(j+1)!. \tag{7.8}$$

If $H_n = u^{n-1}R_n$ and $\widetilde{H}_n = u^{n-1}\widetilde{R}_n$, then

$$\frac{\widetilde{R}_n(0)}{R_n(0)} = -A_n, \qquad A_n = \frac{[n(n-1)]^2}{n(n-1)+2}. \tag{7.9}$$

Proof. The case $n = 1$ is immediate, so assume $n \geq 2$. Every entry in rows $1, \dots, n-1$ of H_n and of $\widetilde{H}_n$ has positive moment index, hence is divisible by u, because every P_j with $j \geq 1$ is (at $\lambda = 0$, $P_1 = 0$ and (5.1) gives $u \mid P_{k+2}$ inductively); factoring u from those rows proves $u^{n-1} \mid H_n$ and $u^{n-1} \mid \widetilde{H}_n$, so R_n and $\widetilde{R}_n$ are polynomials. Every P_j, $j \geq 1$, is divisible by u. Its first coefficient is $q_1 = 0$, $q_2 = -1$, and $q_j = (-1)^{j-1}(j-1)!/2$ for $j \geq 3$, by (5.1). Factor u from rows $1, \dots, n-1$ in H_n and take the constant term of the remaining determinant. The first row then reduces to its top entry, giving

$$[u^{n-1}]H_n = \left(-\frac{1}{2}\right)^m \det[\Gamma(i+j+2) + \delta_{i0}\delta_{j0}]_{i,j=0}^{m-1}.$$

The last matrix is the positive Gram matrix for

$$\mathrm{d}\rho(t) = te^{-t}\,\mathrm{d}t + \delta_0, \qquad t \in [0, \infty), \tag{7.10}$$

so it is nonsingular.

For completeness its determinant and shifted ratio can be computed together. Let $p_j(t) = (-1)^j j! L_j^{(1)}(t)$ be monic Laguerre polynomials for $te^{-t}\,\mathrm{d}t$. Their squared norms are $h_j = j!(j+1)!$, their values at zero are $(-1)^j(j+1)!$, and the coefficient of t^{m-1} in p_m is $-m(m+1)$. For the kernel $K_{m-1}(t,s) = \sum_{j=0}^{m-1} p_j(t)p_j(s)/h_j$,

$$K_{m-1}(0,0) = \frac{m(m+1)}{2}.$$

The matrix determinant lemma gives the factor $1 + K_{m-1}(0,0)$ multiplying the Laguerre determinant, proving (7.8). The monic polynomial orthogonal for (7.10) is

$$Q_m(t) = p_m(t) - \frac{p_m(0)}{1 + K_{m-1}(0,0)} K_{m-1}(t,0).$$

Its orthogonality follows directly from the reproducing property of the kernel. Put $B = m(m+1)$. Its subleading coefficient is $-B + B/(1 + B/2) = -B^2/(B+2)$. The shifted-column determinant formula for that coefficient, together with the extra minus sign from the shifted column in the original q_j, identifies it with $\widetilde{R}_n(0)/R_n(0)$. This proves (7.9). □

Put

$$s_n = \deg R_n = d_n - n + 1 = \left\lfloor \frac{(n-2)^2}{4} \right\rfloor. \tag{7.11}$$

At $\lambda = 0$, the limits of the shifted ratios are

$$b_n = \begin{cases} -n(n-1)/2, & n \text{ even}, \\ -3n(n-1)/4, & n \text{ odd}. \end{cases}$$

Set $\eta_n = A_n + b_n$. Explicitly,

$$\eta_{2k} = \frac{k(k-1)(2k-1)(2k+1)}{k(2k-1)+1}, \qquad \eta_{2k+1} = \frac{k(k-1)(2k+1)(2k+3)}{2[k(2k+1)+1]}. \tag{7.12}$$

These are nonzero for $n \geq 4$. By (7.9),

$$T_n(u) = \frac{\widetilde{R}_n(u) + A_n R_n(u)}{u}$$

is a polynomial of degree $s_n - 1$ with leading coefficient $\ell_n \eta_n$. Define

$$Y_{s_n} = R_n/\ell_n \quad (n \geq 2), \qquad Y_{s_n-1} = T_n/(\ell_n \eta_n) \quad (n \geq 6).$$

The two definitions of Y_0 from $n = 2, 3$ agree and equal one; $Y_1 = u - 21/4$. For $n \geq 6$, $s_{n-1} < s_n - 1 < s_n$, so all further numerator labels are distinct and do not collide with denominator labels. Each assigned Y_j is monic of degree j.

Theorem 7.4. *All formulas of [4, Conjecture 2.7] hold in their stated ranges. A unified form for its diagonal formulas is*

$$\begin{aligned} \alpha_n = {} & \frac{2n^3(n^2+3)}{(n^2-n+2)(n^2+n+2)\zeta} \\ & + \zeta\left(\eta_n \frac{Y_{s_n-1}(u)}{Y_{s_n}(u)} - \eta_{n+1}\frac{Y_{s_{n+1}-1}(u)}{Y_{s_{n+1}}(u)}\right), \qquad n \geq 6. \end{aligned} \tag{7.13}$$

The two off-diagonal formulas are

$$\beta_{2k} = -\frac{2k(2k-1)}{u}\frac{Y_{(k-1)(k-2)}Y_{k(k-1)}}{Y_{(k-1)^2}^2}, \qquad k \geq 1, \tag{7.14}$$

$$\beta_{2k+1} = \frac{1}{u}\frac{Y_{(k-1)^2}Y_{k^2}}{Y_{k(k-1)}^2}, \qquad k \geq 1. \tag{7.15}$$

Proof. The construction gives $\widetilde{H}_n/H_n = -A_n + u\eta_n Y_{s_n-1}/Y_{s_n}$ for $n \geq 6$. Insert this in (7.1) and use

$$A_{n+1} - A_n = \frac{2n^3(n^2+3)}{(n^2-n+2)(n^2+n+2)}$$

to obtain (7.13). Inserting $H_n = u^{n-1}\ell_n Y_{s_n}$ in the second formula of (7.1), and using $\ell_{2k-1}\ell_{2k+1}/\ell_{2k}^2 = 2k(2k-1)$ and $\ell_{2k}\ell_{2k+2}/\ell_{2k+1}^2 = -1$, gives the two β formulas. Substituting $n = 2k$, $n = 4j-1$, and $n = 4j+1$ into (7.13) gives the three diagonal displays in the source, respectively for $k \geq 3$, $j \geq 2$, and $j \geq 2$. □

The source's exclusions of the diagonal indices four and five matter. At $n = 5$, the monic secondary numerator is $u - 177/8$, whereas the already fixed denominator polynomial Y_1 is $u - 21/4$. The stated ranges avoid identifying those different polynomials. At $n = 4$ the secondary numerator is simply $Y_0 = 1$.

7.3 Comparison with the polynomials listed by Lyu and Zhou

Lyu and Zhou list the polynomials X_j and Y_j only at low indices [4, Appendix C]. Table 1 records an exact symbolic comparison of every listed polynomial with the monic constructions used above (X for generic λ, Y at $\lambda = 0$). All eighteen agree exactly, as do the base cases $X_0 = 1$, $X_1 = u - 2\lambda$ and $Y_1 = u - 21/4$.

Family	Index j	Manuscript polynomial	Exact agreement
X_2	2	H_3/ℓ_3 (monic)	yes
X_3	3	F_4 (monic)	yes
X_4	4	H_4/ℓ_4 (monic)	yes
X_5	5	F_5 (monic)	yes
X_6	6	H_5/ℓ_5 (monic)	yes
X_8	8	F_6 (monic)	yes
X_9	9	H_6/ℓ_6 (monic)	yes
Y_2	2	R_5/ℓ_5 (monic)	yes
Y_3	3	T_6 (monic)	yes
Y_4	4	R_6/ℓ_6 (monic)	yes
Y_5	5	T_7 (monic)	yes
Y_6	6	R_7/ℓ_7 (monic)	yes
Y_8	8	T_8 (monic)	yes
Y_9	9	R_8/ℓ_8 (monic)	yes
Y_{11}	11	T_9 (monic)	yes
Y_{12}	12	R_9/ℓ_9 (monic)	yes
Y_{15}	15	T_{10} (monic)	yes
Y_{16}	16	R_{10}/ℓ_{10} (monic)	yes

Table 1: Exact agreement of the manuscript's polynomials with those listed in [4, Appendix C].

8 Consequences and limits

8.1 Rationality in special-frequency existence

The rationality assumption in the all-degree existence theorem of [2, Theorem 2.2] cannot be removed throughout the real parameter interval.

Proposition 8.1. *For every integer $m \geq 2$, there is an irrational $\lambda_m > 1/2$ such that, with $\zeta_m = j_{\lambda_m - 1, m}$,*

$$D_3^{(\lambda_m - 1/2)}(\zeta_m) = 0.$$

There are infinitely many distinct such parameters. At each of these points the monic orthogonal polynomial of degree three does not exist, although every even degree exists uniquely.

Proof. The recurrence (5.1) gives

$$H_3(u;\lambda) = 2(2\lambda+1)[u^2 - \lambda(2\lambda+5)u + 4\lambda^2].$$

For $\lambda \geq 1/2$, set

$$F_m(\lambda) = j_{\lambda-1,m}^4 - \lambda(2\lambda+5)j_{\lambda-1,m}^2 + 4\lambda^2.$$

Positive Bessel zeros vary continuously with the order in this range. At $\lambda = 1/2$, put $u_m = (m-1/2)^2\pi^2$. Then $F_m(1/2) = u_m^2 - 3u_m + 1 > 0$ for $m \geq 2$. The fixed-index large-order expansion $j_{\nu,m} = \nu + O(\nu^{1/3})$ [14, §10.21, Eq. 10.21.32] gives $F_m(\lambda)/\lambda^4 \to -1$. The intermediate value theorem therefore gives a root $\lambda_m > 1/2$.

At a positive zero of $J_{\lambda-1}$, the function J_λ is nonzero: a common zero would also make the derivative of $J_{\lambda-1}$ vanish at an ordinary point of its differential equation. Thus (5.9) converts

$F_m(\lambda_m) = 0$ into the asserted determinant zero. If λ_m were rational, [2, Theorem 2.2] would give nonvanishing of every determinant there. Hence λ_m is irrational. A fixed parameter permits at most two positive u solving the displayed quadratic, so finitely many parameters cannot account for all indices $m \geq 2$.

Finally, let p_2 be the monic quadratic, which exists because $D_2 > 0$. At $D_3 = 0$, its norm is zero, so it is orthogonal through degree two. If a monic orthogonal cubic existed, symmetry of the bilinear form would also make p_2 orthogonal to degree three. The size-four moment matrix would then be singular, contrary to $D_4 > 0$. The even-degree conclusion follows from Theorem 1.1. □

This excludes a natural unrestricted-real-parameter extension of the 2008 existence theorem; no published conjecture of that extension is being attributed to its authors.

8.2 Symmetry is insufficient

Consider the symmetric positive probability measure

$$\nu = \frac{1}{6}(\delta_{-1} + \delta_{-1/2} + 2\delta_0 + \delta_{1/2} + \delta_1).$$

Form its Fourier Hankel determinants using $\mu_j = \int t^j e^{\mathrm{i}xt}\,\mathrm{d}\nu(t)$. At $x = 2\pi$, odd moments vanish and

$$\mu_0 = \frac{1}{3}, \qquad \mu_2 = \frac{1}{4}, \qquad \mu_4 = \frac{5}{16}, \qquad \mu_6 = \frac{21}{64}.$$

Thus $D_2 = 1/12 > 0$, whereas parity-block decomposition gives

$$D_4 = \left(\frac{1}{3}\frac{5}{16} - \frac{1}{16}\right)\left(\frac{1}{4}\frac{21}{64} - \frac{25}{256}\right) = -\frac{1}{1536} < 0. \tag{8.1}$$

The Gegenbauer structure in the positivity theorem is substantive. Positive-definite functions may take negative values, so this example does not conflict with the Fourier-Wronskian positive-definiteness results of [3].

8.3 Spherical-Bessel determinants and critical length

Define

$$f_n(x) = \sqrt{\frac{\pi}{2}}x^{n+1/2}J_{n+1/2}(x), \qquad W_k[f](x) = (-1)^{k(k-1)/2}\det[f^{(i+j)}(x)]_{i,j=0}^{k-1}. \tag{8.2}$$

Appendix A proves that, for $1 \leq k \leq n+1$,

$$W_k[f_n](x) = \frac{2^{k(k-n-2)}(n!)^k}{\prod_{q=n+1-k}^{n}(q!)^2}x^{k(2n+2-k)}D_k^{(n+1-k)}(x). \tag{8.3}$$

All constants are positive. Theorem 1.1 therefore gives

Corollary 8.2. *For every $n \geq 0$, every even k with $2 \leq k \leq n+1$, and every $x > 0$, one has $W_k[f_n](x) > 0$.*

Let

$$U_n = \operatorname{span}\{x^j \cos x, x^j \sin x : 0 \le j \le n\}.$$

Its critical length is the supremum of interval lengths on which every nonzero member has fewer than $2n+2$ zeros, counted with multiplicity. Carnicer, Mainar and Peña [10] conjecture that this length is the first positive zero $j_{n+1/2,1}$ of $J_{n+1/2}$.

Corollary 8.3. *The critical length of U_3 is $j_{7/2,1}$.*

Proof. The space U_n is invariant under translation and reflection, and $u_j = f_n^{(2n+1-j)}$, $0 \le j \le 2n+1$, is a canonical basis at zero. Here canonical means $u_j^{(i)}(0) = 0$ for $i < j$ and $u_j^{(j)}(0) \neq 0$, as in [10, Eq. (2)]. Indeed f_n has a zero of order $2n+1$ there and belongs to $\ker(D^2+1)^{n+1}$. The critical length is finite: f_n has $2n+1$ zeros at the origin, counted with multiplicity, and another at its first positive zero. Proposition 3 of [10] states that the critical length is the least positive zero among the suffix Wronskians with $j > (2n+1)/2$. Such a suffix has size $k = 2n+2-j$, so precisely $1 \le k \le n+1$ occurs. Reversing its columns identifies its Wronskian with $W_k[f_n]$.

For $n = 3$, only $k = 1, 2, 3, 4$ is required. The function $W_1 = f_3$ is positive on $(0, j_{7/2,1})$ and vanishes at its right endpoint. Corollary 8.2 gives positivity of W_2, W_4 for every $x > 0$. Kounchev and Render [11, Corollary 21] (an unrefereed preprint, on which this corollary depends) prove that $W_3[f_3]$ is positive on $(0, j_{7/2,1})$. Thus none of the required suffix Wronskians vanishes earlier, and one vanishes at the stated endpoint. □

For general n, (8.3) leaves a first-zero comparison for odd determinants of size at least five. Infinite oscillation and simple zeros do not supply that comparison. No resolution of the general critical-length conjecture is claimed.

9 Checks and the scope of the literature comparison

At $\alpha = 0$ and size two,

$$D_2^{(0)}(x) = \frac{4(x^2 - \sin^2 x)}{x^4}, \qquad D_2^{(0)}(0) = \frac{4}{3}.$$

This checks the sign and $C_{1,0} = 4$ in (3.6). The general coefficient formula gives $C_{2,0} = 256$, consistent with the exponent-zero endpoint calculation in [1]. At odd size one, $D_1^{(0)}(x) = 2\sin x/x$ and $T_{0,0} = 0$. Exact symbolic calculations from (5.1) for $n = 1, \dots, 5$ give degrees $0, 1, 2, 4, 6$ and leading coefficients

$$1, \quad -1, \quad 2(2\lambda+1), \quad 4(2\lambda+1)^2, \quad 32(2\lambda+1)^3(2\lambda+3).$$

They verify normalisation and signs; the all-rank arguments above are independent of these finite checks.

The companion script `verify.py` records its results in `verification_results.json`. At 90 decimal digits it checked 120 even determinants: sizes $2, 4, 6, 8$, frequencies $0, 0.1, \pi, 10, 50$, and exponents $-0.99, -0.5, 0, 0.5, 1.4142135623730950488, 3$. All computed determinants were positive. The moments were cross-checked against an independent hypergeometric formula. The same script checks (2.4), the exact $D_2^{(0)}$ identity, the tail constants, and the Toda identity. Its exact symbolic checks cover H_n through $n = 5$, generic shifted numerators through $n = 4$, the general-parameter residual polynomials through $n = 5$, and the zero-parameter reduced determinants and shifted ratios

through $n = 9$. It also verifies both exceptional polynomials (6.10)–(6.11) and the atomic example (8.1). These computations are checks of formulas and normalisations, not a numerical proof of the theorems.

The comparison made for this draft includes the full 2008 paper, the 2022 exponent-zero existence proof, Dimitrov–Xu's Fourier-Wronskian results, the 2025 rank-three critical-length result, and Lyu–Zhou's 2026 recurrence paper. The finite verification record [15] reports degree checks through $n = 16$ and more limited recurrence-coefficient checks. Such a finite verification is different in scope from the all-rank arguments here.

A close classical comparison is also important. The Karlin–Szegő even-determinant inequalities, as set out in [9, Section 2.2], differentiate $u^{-\nu/2}J_\nu(2\sqrt{u})$ with respect to u and hence lead to order-shifted Bessel entries. Our determinant differentiates $x^{-\nu}J_\nu(x)$ with respect to x. A nonlinear substitution does not generally preserve derivative Hankel determinants: writing $\mathcal{H}_2[f] = ff'' - (f')^2$, the chain rule gives

$$\mathcal{H}_2[f \circ \phi] = \phi'^2(\mathcal{H}_2[f] \circ \phi) + \phi''((ff') \circ \phi).$$

Thus that classical result is not an immediate reparametrisation of Theorem 1.1. This comparison rules out the direct identification; it does not claim to exclude every deeper connection.

The earlier claim. Ashfaque [16] (ResearchGate, 24 July 2026) states in its abstract and Proposition 3.1 that the symbolic recurrence formulas conjectured by Lyu and Zhou follow from a discrete Painlevé IV and tau-function identification. Our description is based on a full reading of that six-page document on 2 November 2026; the host later blocked automated access, so readers should consult the original. In its notation the weight is $(1 - x^2)^{\lambda-1/2}e^{\mathrm{i}\zeta x}$ with $J_{\lambda-1}(\zeta) = 0$, the recurrence is $xP_n = P_{n+1} + \mathrm{i}\alpha_n P_n + \beta_n P_{n-1}$ with real α_n, and it sets $u_n = \zeta\alpha_n - (2n + 2\lambda - 1)$. Proposition 3.1 defines $D_n^{(1)} = \det[\mu_{i+j+1}]_{i,j=0}^{n-1}$, shifting every moment index, and asserts

$$u_n = \zeta\frac{D_n^{(1)}}{D_n} - (2n + 2\lambda - 1). \tag{9.1}$$

This cannot hold as printed. For real ζ, $\overline{\mu_j} = (-1)^j\mu_j$, so the argument used in Section 4 shows that $D_n^{(1)}$ is purely imaginary when n is odd, while D_n is real. At $n = 1$, $\mu_1/\mu_0 = \mathrm{i}J_{\lambda+1}(\zeta)/J_\lambda(\zeta) = 2\mathrm{i}\lambda/\zeta$ at every nonzero real zero of $J_{\lambda-1}$ (where $J_\lambda(\zeta) \neq 0$, since adjacent Bessel functions have no common nonzero zero), by the three-term Bessel recurrence. The right side of (9.1) is then $2\mathrm{i}\lambda - (2\lambda + 1)$, which is not real when $\lambda \neq 0$, whereas the left side is real. At $\lambda = 0$ the two sides are -1 and 0 respectively, using Lyu and Zhou's initial value $\alpha_1 = 1/\zeta$. The failure is not an index slip: the right side alternates between imaginary and real as the parity of n changes, so no fixed index shift or constant phase repairs it for all n. Moreover the subleading coefficient of a monic orthogonal polynomial comes from a single last-column shift, and α_n is a difference of two such ratios. A corrected argument along those lines may exist; we make no judgement about one. The arguments in this manuscript do not use that proposition or the claimed resolution, but the earlier public claim is part of the priority record and we do not claim to be first to assert these formulas.

This is a bounded comparison, not an exhaustive priority determination. In particular, no claim of historical firstness is inferred solely from the absence of a located valid proof. The central mathematical claims of the draft are the extension in Theorem 1.1, the explicit all-rank algebraic results in Theorems 5.1 and 6.1, and the stated critical-length corollary. They are provided with proofs so that their validity and their relation to earlier work can be assessed separately.

A An exact Bessel–Jacobi determinant identity

This appendix proves (8.3), including its constant. The two standard inputs are the normalised unitary HCIZ integral [12, Eq. (2)] and the Jacobi law for random projection compressions [13, Theorem 2.2]. The determinant reduction between them is given explicitly.

A.1 A confluent orbital-integral identity

Let $1 \le k \le N$ and initially let $a_1, \dots, a_N$ be distinct complex numbers. Put

$$p(z) = \prod_{j=1}^{N} (z - a_j), \qquad g(x) = \sum_{j=1}^{N} \frac{e^{a_j x}}{p'(a_j)}, \qquad A = \operatorname{diag}(a_1, \dots, a_N), \quad B = \operatorname{diag}(0_{N-k}, 1_k).$$

Normalise Haar measure on $U(N)$ to mass one and set

$$I_{N,k}(A; x) = \int_{U(N)} e^{x \operatorname{tr}(AUBU^*)} \, \mathrm{d}U, \qquad Q_{N,k} = \frac{\prod_{j=N-k}^{N-1} j!}{\prod_{j=0}^{k-1} j!}.$$

We claim that

$$I_{N,k}(A; x) = Q_{N,k} x^{-k(N-k)} W_k[g](x). \tag{A.1}$$

Start with the normalised HCIZ formula

$$\int_{U(N)} e^{x \operatorname{tr}(AUCU^*)} \, \mathrm{d}U = \left(\prod_{j=1}^{N-1} j! \right) \frac{\det[e^{x a_i c_j}]_{i,j=1}^{N}}{x^{N(N-1)/2} \Delta(a) \Delta(c)}. \tag{A.2}$$

Analyticity extends the usual real diagonal formula to complex diagonal entries. Coalesce $N - k$ eigenvalues of C at zero and k at one. The confluent columns are

$$\frac{(x a_i)^j}{j!} \quad (0 \le j < N - k), \qquad \frac{(x a_i)^\ell e^{x a_i}}{\ell!} \quad (0 \le \ell < k).$$

The cross-cluster Vandermonde factors are one. Factoring powers and factorials gives

$$I_{N,k}(A; x) = Q_{N,k} x^{-k(N-k)} \frac{\det M(x)}{\Delta(a)}, \quad M = [a_i^j \ (0 \le j < N - k) \mid a_i^\ell e^{x a_i} \ (0 \le \ell < k)]. \tag{A.3}$$

Let $V = [a_i^j]_{1 \le i \le N,\, 0 \le j < N}$. The first $N - k$ columns of $V^{-1} M$ are coordinate vectors, so its determinant is that of its bottom-right k-by-k block. In a column labelled by ℓ, its entries are coefficients of the polynomial interpolating $a_i^\ell e^{x a_i}$ at the nodes. The coefficient of z^{N-1-s} in the ith Lagrange polynomial is

$$\frac{(-1)^s e_s(a_1, \dots, \widehat{a_i}, \dots, a_N)}{p'(a_i)} = \frac{a_i^s + \sum_{q=0}^{s-1} c_{s,q} a_i^q}{p'(a_i)},$$

where the $c_{s,q}$ depend only on the full node set. The corresponding entry is therefore $g^{(s+\ell)} + \sum_{q<s} c_{s,q} g^{(q+\ell)}$. The bottom rows have $s = k-1, \dots, 0$. Reverse them, producing $(-1)^{k(k-1)/2}$, and eliminate the lower triangular terms by unit triangular row operations. The resulting determinant is exactly $W_k[g]$. This proves (A.1).

The formulas extend continuously to coalescing nodes. For example, g is the exponential divided difference, equivalently

$$g(x) = \frac{1}{2\pi \mathrm{i}} \oint \frac{e^{xz}}{p(z)} \, \mathrm{d}z,$$

where the contour encloses all roots of p.

A.2 Coalescing at the spherical-Bessel spectrum

Take $N = 2n+2$, $m = n+1$, and $A = \operatorname{diag}(-\mathrm{i}I_m, \mathrm{i}I_m)$. Then $p(z) = (z^2+1)^{n+1}$ and g is the fundamental solution of

$$(D^2+1)^{n+1} g = 0, \qquad g^{(j)}(0) = 0 \ (0 \le j < 2n+1), \qquad g^{(2n+1)}(0) = 1.$$

The spherical-Bessel function f_n belongs to this solution space and satisfies

$$f_n(x) = \frac{x^{2n+1}}{(2n+1)!!} + O(x^{2n+3}), \qquad \frac{(2n+1)!}{(2n+1)!!} = 2^n n!.$$

Hence $g = f_n/(2^n n!)$, and (A.1) becomes

$$I_{N,k}(A;x) = \frac{Q_{N,k}}{(2^n n!)^k} x^{-k(N-k)} W_k[f_n](x). \tag{A.4}$$

Write $A = \mathrm{i}(2P - I_N)$, where P is a rank-m orthogonal projection. If V consists of the k columns of U selected by B, then $\operatorname{tr}(AUBU^*) = \mathrm{i}(2\operatorname{tr}(V^*PV) - k)$. For $k \le m$, the eigenvalues $s_1, \dots, s_k$ of V^*PV have Jacobi density

$$\text{constant} \cdot \Delta(s)^2 \prod_{j=1}^{k} s_j^{m-k} (1-s_j)^{m-k}, \qquad 0 < s_j < 1. \tag{A.5}$$

After $t_j = 2s_j - 1$, this is proportional to $\Delta(t)^2 \prod (1-t_j^2)^{m-k}$. Andréief's identity therefore identifies the same orbital integral as

$$I_{N,k}(A;x) = \frac{D_k^{(n+1-k)}(x)}{D_k^{(n+1-k)}(0)}. \tag{A.6}$$

Finally, (2.4) gives

$$Q_{2n+2,k} D_k^{(n+1-k)}(0) = 2^{k(2n+2-k)} \prod_{q=n+1-k}^{n} (q!)^2.$$

Combining (A.4) and (A.6) proves (8.3). Both sides are entire, so the identity extends to complex frequency, including zero by continuation.

For $k = 1$ the identity reduces to $f_n = x^{2n+1} D_1^{(n)}/(2^{n+1} n!)$. For $n = 1, k = 2$, it gives $W_2[f_1] = x^2 - \sin^2 x$, agreeing with $D_2^{(0)} = 4(x^2 - \sin^2 x)/x^4$ and the constant $1/4$.

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